

Stem Cell Growth using Mycelium Framework

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Abstract

Mycelium

Keywords: Place text here

1. INTRODUCTION

Using mycelium scaffold to influence stem cell adaptation. Creating the stem cell niche by growing the stem cell in a natural biological scaffold.

2. LITERATURE REVIEW

In [1], the affects of a biological gause on wound healing is observed. It concludes that the mycelium biomaterials are a promising approach to alternative accelrative wound healing. Both tested materials, *G. lucidum* and *P. ostreatus* enhance collagen I gene expression. *G. lucidum* performs marginally better. The β -glucans in the mycelia are crucial in prompting the body to synthesize types I and III collagen.

In [2] it is demonstrated that the mycelia of *Pleurotus ostreatus* can be directly used as the scaffold for cell growth. It had good viability, and morphology compared to the control samples used in [2]. Human cells were directly plated onto the un-modified and non-functionalised mycelial scaffold. Worth noting; even after Hyphal cell deactivation, *Ganoderma lucidum* was not fully compatible for direct plating. Its water extract contained concentrations of organic acids (Ganoderic acid V and Oleandrin). This was detrimental for HDFa cells survival

In [3], the stem cell niche is discussed. It details the effect of the microenvironment on stem cell differentiation. It provides a methodology for defining the cell types that stem cells can grow into, and the condiitons that are needed for this to occur. Under homeostatic conditions, there is a symbiotic relationship between the stem cells and

their niche. What must be further investigated is the response of the stem cells and their niche to tissue injury during the disease progression and aging. Further study must be done on genetic factors that influence the stem cell niche.

In [4] a computational model is developed for identifying niche factors that are used in the rejuvenation of quiescent stem cells. Initally, the study focused on skeletal stem cells, the stem like cells do not degrade till much later life. This model is used can be used for the identification of niche factors of multiple different tissue types.

3. INTRODUCTION

Let me be frank, we don't see the full picture and we should look at the effect of fungi on our systems and Whether they can be modified to help us. a clear example would be enabling stem cells to grow within modified mycelial compounds.

3.1. Example Subsection

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3.2. Example Second Subsection

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3.2.1. Example Subsubsection

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4. RESULTS AND DISCUSSION

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5. CONCLUSION AND FUTURE WORK

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6. DECLARATION

Conflict of Interest: Place text here

Author Contribution: Place text here

7. REFERENCES

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